Note: The major questions on this topic is Write the Principle or/and applications of _____Spectroscopy.

I. UV Visible spectroscopy:

Principle:

- The UV Visible Radiation (200nm-800nm) Possesses certain energy to excite the valence electrons of the molecules in the sample from its ground state to its excited state, during which a certain amount of energy is been absorbed by the molecule in the sample .
- UV Spectrum is obtained by measuring the light absorbed or transmitted by/from the sample as a function of wavelength in comparison to a reference or blank sample using a detector.
- Quantitative measurements in UV Visible spectroscopy is evaluated using Beer-Lamberts law which states that "For a given material, the sample path length and concentration of the sample are directly proportional to the absorbance of the light."
    - A=Ɛ .C.L Where,
        - Ɛ (epsilon)- Molar absorption coefficient
        - A- Absorbance
        - C- concentration of the sample
        - L- Path length (width of cuvette)

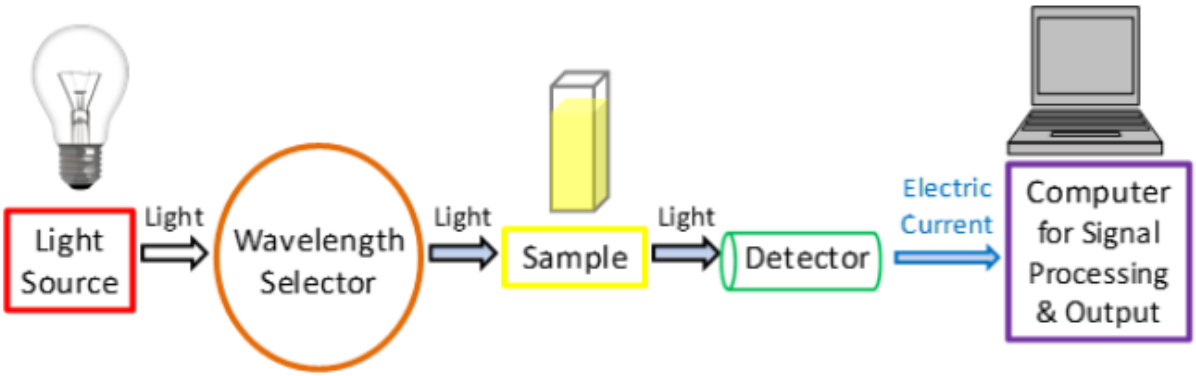


FIG: A simplified schematic representation of the main components in a UV-Vis

Note: This representation is only for understanding purpose, you need not include the diagram in principle part.

Applications:

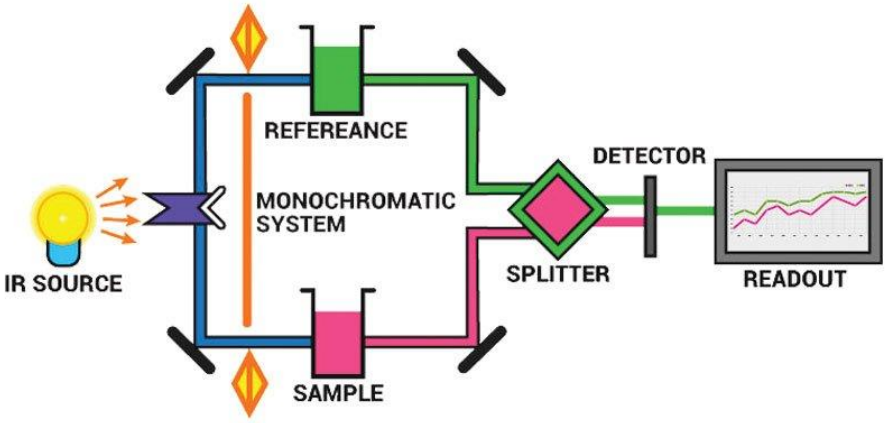
1. DNA and RNA analysis
   - Verifying the purity and concentration of RNA and DNA is one particularly widespread application.
   - When preparing DNA or RNA samples, for example for downstream applications such as sequencing, it is often important to verify that there is no contamination of one with the other, or with protein or chemicals carried over from the isolation process.
   - *The Nucleic acid samples show maximum absorbance at 260 nm*
2. Pharmaceutical analysis
   - One of the most common uses of UV-Vis spectroscopy is in the industry. In particular, processing UV-Vis spectra using mathematical derivatives allows overlapping absorbance peaks in the original spectra to be resolved to identify individual pharmaceutical compounds
   - For example, benzocaine, a local anesthetic, and chlortetracycline, an antibiotic, can be identified simultaneously in commercial veterinary powder formulations by applying the first mathematical derivative to the absorbance spectra.
3. Monitoring the growth in Bacterial culture
   - UV-Vis spectroscopy is often used in bacterial culturing.
   - OD measurements are routinely and quickly taken using a wavelength of 600 nm to estimate the cell concentration and to track growth.
   - 600 nm is commonly used and preferred due to the optical properties of bacterial culture media in which they are grown and to avoid damaging the cells in cases where they are required for continued experimentation.
4. In Beverage analysis
   - The identification of particular compounds in drinks is another common application of UV-Vis spectroscopy.
   - Caffeine content must be within certain legal limits, for which UV light can facilitate quantification.
   - Certain classes of colored substances, such as anthocyanin found in blueberries, raspberries, blackberries, and cherries, are easily identified by matching their known peak absorbance wavelengths in wine for quality control using UV-Vis absorbance.
5. Measuring the absorbance of hemoglobin to determine hemoglobin concentrations may be used in cancer research
6. In wastewater treatments, UV-Vis spectroscopy can be used in kinetic and monitoring studies to ensure certain dyes or dye by-products have been removed properly by comparing their spectra over time

II. <u>Infra-red spectroscopy</u>:

Principle:

- IR radiation (800nm to 1mm) is passed through the sample under study. When the frequency of a specific vibration is equal to the frequency of the IR radiation directed on the molecule, the molecule absorbs the radiation.

- This causes vibrational motion within the molecules (vibrations can be stretching or bending)
- The transmitted light (% transmittance) after the absorbance is detected by a detector and an IR spectrum is obtained, Which reveals how much energy was absorbed at each frequency.



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Applications of IR Spectroscopy

Infrared spectroscopy is widely used in industry as well as in research. It is a simple and reliable technique for measurement, quality control and dynamic measurement. It is also employed in forensic analysis in civil and criminal analysis.

Some of the major applications of IR spectroscopy are as follows:

1. Identification of functional group and structure elucidation

- Entire IR region is divided into group frequency region and fingerprint region. Range of group frequency is 4000-1500 cm^{-1}. While that of finger print region is 1500-400 cm^{-1}.
- In group frequency region, the peaks corresponding to different functional groups can be observed. According to corresponding peaks, functional group can be determined.
- Each atom of the molecule is connected by bond and each bond requires different IR region so characteristic peaks are observed. This region of IR spectrum is called as finger print region of the molecule. It can be determined by characteristic peaks.

2. Identification of substances

- IR spectroscopy is used to establish whether a given sample of an organic substance is identical with another or not.

- This is because large number of absorption bands is observed in the IR spectra of organic molecules and the probability that any two compounds will produce identical spectra is almost zero.
- So if two compounds have identical IR spectra then both of them must be samples of the same substances.

3. Studying the progress of the reaction

- Progress of chemical reaction can be determined by examining the small portion of the reaction mixture withdrawn from time to time. The rate of disappearance of a characteristic absorption band of the reactant group and/or the rate of appearance of the characteristic absorption band of the product group due to formation of product is observed.

4. Detection of impurities

- IR spectrum of the test sample to be determined is compared with the standard compound. If any additional peaks are observed in the IR spectrum, then it is due to impurities present in the compound.

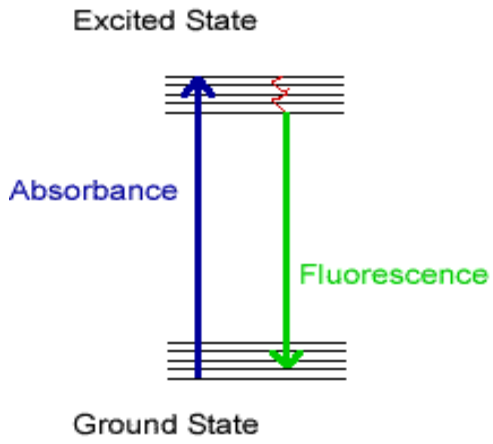
5. Quantitative analysis

- The quantity of the substance can be determined either in pure form or as a mixture of two or more compounds. In this, characteristic peak corresponding to the drug substance is chosen and log I0/It of peaks for standard and test sample is compared. This is called base line technique to determine the quantity of the substance.

III. Fluorescence spectroscopy

Principle

- Molecules have various states referred to as energy levels.
- The species being examined has a ground electronic state (a low energy state), and an excited electronic state of higher energy. Within each of these electronic states there are various vibrational states.
- In fluorescence, the species is first excited, by absorbing a photon, from its ground electronic state to one of the various vibrational states in the excited electronic state.
- The molecule then drops down from its higher energy state .i.e., from excited electronic state to the ground electronic state again, emitting a photon in the process.
- The emitted photons will have different energies, and thus frequencies. Therefore, by analyzing the different frequencies of light emitted in fluorescent spectroscopy, along with their relative intensities, the structure of the different vibrational levels can be determined.



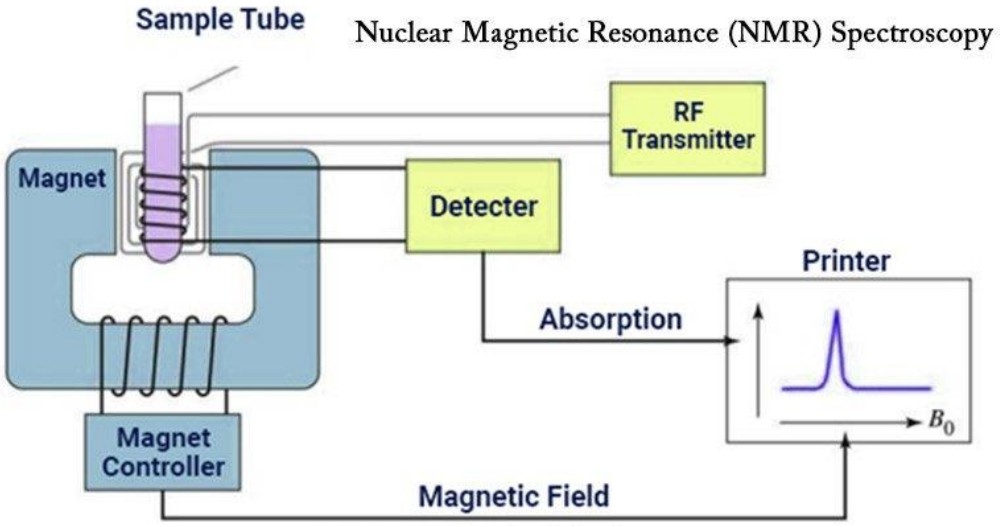
Applications:

1. Fluorescence spectroscopy is used in, among others, biochemical, medical, and chemical research fields for analyzing organic compounds.
2. Its use in differentiating malignant skin tumors from benign.
3. Fluorescence spectroscopy can be adapted to the microscopic level using microfluorimetry
4. In the field of water research, fluorescence spectroscopy can be used to monitor water quality by detecting organic pollutants
5. Advance fluorescence spectroscopy can be used in detection of bacterial contamination of water

IV. NMR Spectroscopy (NMR- Nuclear Magnetic Resonance)

Principle

- The principle behind NMR is that many nuclei have spin and all nuclei are electrically charged.
- If an external magnetic field is applied, an energy transfer is possible between the base energy to a higher energy level (generally a single energy gap).
- The energy transfer takes place at a wavelength that corresponds to radio frequencies
- And when the spin returns to its base level, energy is emitted at the same frequency.
- The signal that matches this transfer is measured in many ways and processed in order to yield an NMR spectrum for the nucleus concerned.



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Applications

1. Chemistry laboratories/industries
   - analyst rely on NMR Spectroscopy as a tool to chart the complex molecular structures of matter
2. Food quality control and research
   - Before hitting the consumer market, foods undergo strict quality control and research studies.
   - NMR Spectroscopy is widely used across the industry to map protein structures, profile amino acids, identify carotenoids and quantify metabolites.
3. Identifying human disorders
   - NMR Spectroscopy is an invaluable tool for identifying human disorders.
   - The cellular metabolism contains important biomarkers used to diagnose a wide range of conditions, including tuberculosis, pneumonia and malaria.
   - NMR Spectroscopy allows researchers to identify these tell-tale biomarkers and treat patients accordingly.
4. Cancer diagnosis
   - Over the past few decades, NMR Spectroscopy has become a key technique for cancer diagnosis.
   - The ability to analyze abnormal behavior in the cellular metabolism allows scientists to detect the metabolite-based biomarkers associated with cancers.
5. Environmental monitoring

- Spectrometers are used to detect and characterize contaminants in air, soil and water samples as well as monitor the metabolic responses of organisms exposed to these contaminants.

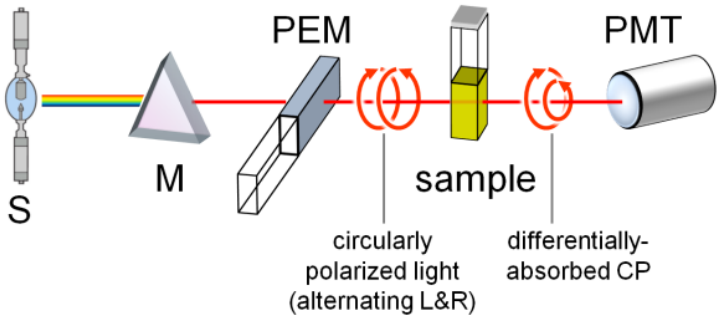
6. Drug discovery and development
   - NMR Spectroscopy is used in the drug discovery and development researches.

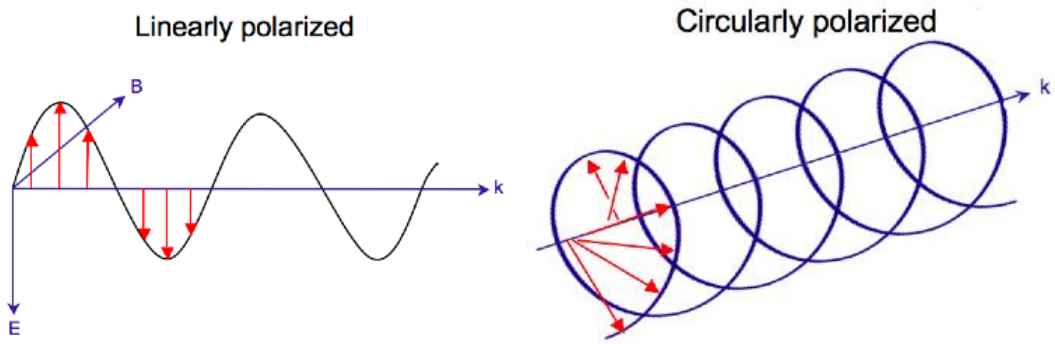
V. Circular Dichroism

Principle

When plane-polarized light passes through an optically active medium that has an absorption peak in the ultraviolet region, the left-handed and right-handed circularly polarized light components contained in the region have different propagation speeds (due to different refractive indexes) and different intensities. This phenomenon is circular dichroism (CD).

- When plane circularly polarized light passes through the compound under study, which contains the chromophores.
  (for ex: In protein and polypeptide molecules, peptide bonds, aromatic amino acid residues and disulfide bridges in the peptide chain backbone are the main photoactive chromophores.)
- These chromophores absorb left and right circularly polarized light differently, resulting in a difference in the amplitude of the polarized light vector turning the circularly polarized light into elliptically polarized light which produces the circular dichroism spectrum.
- The **circular dichroic spectrum** is developed on the basis of the elipticity.
- The **circular dichroic spectrum** is a difference spectrum that shows the difference of the absorption spectrum of the sample under the irradiation of left and right polarized light.





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Applications:

- It is used to investigate the secondary structure of proteins that cannot be crystallized
- It is used to investigate charge transfer transitions
- It is used to examine geometric and electronic structure by probing metal transitions
- It is used to investigate the effect of drug binding on protein secondary structure
- It is used to study the ligand induced conformational change